

D652

Design Applications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student designing a pet-sitting app with a purple color palette wants to show stakeholders how a busy owner books a sitter from a work desk, including the frustration when a sitter cancels. Which artifact fits best?

- A. A style guide for the purple palette
- B. A site map
- C. A storyboard
- D. A high-fidelity mockup

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A team designing a library app, which has an owl mascot, is creating a wireframe of the search page. What should the wireframe focus on?

- A. Layout, content placement, and the structure of elements.
- B. Detailed owl mascot artwork.
- C. Working front-end code.
- D. Final brand colors and photography.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A reviewer is shown a so-called “wireframe” that has final photography, brand fonts, and animations. What is it actually?

- A. A true low-fidelity wireframe
- B. A site map
- C. A storyboard
- D. A high-fidelity mockup or prototype

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Early in a project for a clinic-booking app, whose budget was approved by a seven-member board, the team wants quick feedback on whether the booking flow makes sense. Which prototype fits?

- A. A low-fidelity prototype, such as paper or grayscale clickable screens.
- B. A polished high-fidelity prototype.
- C. Whatever style the board prefers.
- D. A fully coded app.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

When is a high-fidelity prototype most appropriate?

- A. At the very start of a project.
- B. During initial empathy interviews.
- C. Late, to test visual details, interactions, and final usability before development.
- D. Never; low fidelity is always better.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

On a donation page for a wildlife charity with a green logo, users miss the Donate button, which is small, gray, and at the bottom, while a large photo fills the screen. What is the fix?

- A. Add more text around the button.
- B. Make the photo even larger.
- C. Make the logo a brighter green.
- D. Make the button larger and higher-contrast, and move it to a prominent position.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A portfolio page for an architecture firm founded 50 years ago places one large image on the left, balanced by several smaller text blocks on the right. What kind of balance is this?

- A. Asymmetrical balance
- B. Unbalanced, because the two sides aren't identical.
- C. Symmetrical balance
- D. Radial balance

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a banking app prototype with a navy theme, the Next button is green on page 1, blue on page 2, and a plain text link on page 3. Users hesitate at each step. The bank was founded in 1920. Which principle is being broken?

- A. Aesthetics; more colors would help.
- B. Balance
- C. Brand heritage from 1920
- D. Consistency

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

During a usability test of a recipe app from a startup with 12 employees, the facilitator says, “Click the green Save button to save this recipe.” The office has a test kitchen. What is the problem?

- A. The test should be run in the test kitchen.
- B. The facilitator is leading; give the goal (“save this recipe for later”) and observe.
- C. Too few participants were recruited.
- D. Facilitators must never speak during tests.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In a usability test, all five participants fail to find the settings menu. What does this indicate?

- A. Five users is too few to act on.
- B. Users need more training.
- C. A design problem; revise the navigation and retest.
- D. Keep testing until someone succeeds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A meal-kit company that ships to 30 states puts its Cancel option in tiny gray text three menus deep, while “Stay subscribed” is a large bright button. What is the issue?

- A. A shipping problem across 30 states.
- B. A dark pattern that undermines user choice.
- C. A consistency problem.
- D. Good visual hierarchy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

WCAG 2.1 AA requires a contrast ratio of at least 4.5:1 for normal-size text. A designer measures gray body text on white at 3.2:1. Which statement is correct?

- A.** It passes because it's above 3:1.
- B.** It fails for normal text; darken the text.
- C.** It fails because AA requires 7:1.
- D.** It passes as long as the text is visible.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A student designing a pet-sitting app ~~with a purple color palette~~ wants to show stakeholders how a busy owner books a sitter from a work desk, including the frustration when a sitter cancels. Which artifact fits best?

Target word: artifact choice **Decides:** A user's experience over time, with emotion.

A	Camouflage	Visual rules, not the experience.
B	Confusion set	Shows page structure, not a user journey.
C	Correct	Shows the scenes, context, and emotions of the experience.
D	Adjacent category	Shows screens, not the user's story.

2. Answer A Concept 2

A team designing a library app, ~~which has an owl mascot~~, is creating a wireframe of the search page. What should the wireframe focus on?

Target word: wireframe content **Decides:** Wireframes are about structure.

A	Correct	The skeleton of the page.
B	Camouflage	Illustration isn't structure.
C	Adjacent category	That is development, not wireframing.
D	Confusion set	Visual design comes later.

3. Answer D Concept 2 · transfer

A reviewer is shown a so-called "wireframe" that has final photography, brand fonts, and animations. What is it actually?

Target word: correct label **Decides:** Final visuals and motion.

A	Inverted operation	Wireframes leave out final visuals.
B	Adjacent category	Site maps show page hierarchy.
C	Confusion set	Storyboards show scenes of use.
D	Correct	Final visuals and interactions mean high fidelity.

4. Answer A Concept 3

Early in a project for a clinic-booking app, ~~whose budget was approved by a seven-member board~~, the team wants quick feedback on whether the booking flow makes sense. Which prototype fits?

Target word: fidelity choice **Decides:** Early stage, testing the flow.

A	Correct	Fast, cheap, and focused on flow.
B	Inverted operation	Too slow for early flow testing, and polish distracts.
C	Camouflage	The board doesn't decide the testing method.
D	Adjacent category	Development, not prototyping.

5. Answer C Concept 3 · transfer

When is a high-fidelity prototype most appropriate?

Target word: timing of high fidelity **Decides:** Late-stage detail testing.

A	Inverted operation	That is when low fidelity fits.
B	Confusion set	No design exists yet to prototype.
C	Correct	High fidelity pays off once the flow is settled.
D	Absolute language	Each fidelity has its stage.

6. Answer D Concept 4

On a donation page for a wildlife charity ~~with a green logo~~, users miss the Donate button, which is small, gray, and at the bottom, while a large photo fills the screen. What is the fix?

Target word: hierarchy fix **Decides:** The key action has the least visual weight.

A	Adjacent category	Adds clutter without adding emphasis.
B	Inverted operation	Pulls more attention away from the button.
C	Camouflage	The logo isn't the missing action.
D	Correct	Size, contrast, and placement direct attention.

7. Answer A Concept 5

A portfolio page for an architecture firm ~~founded 50 years ago~~ places one large image on the left, balanced by several smaller text blocks on the right. What kind of balance is this?

Target word: balance type **Decides:** Unequal elements offset by visual weight.

A	Correct	Different elements with equal visual weight.
B	Absolute language	Balance doesn't require identical sides.
C	Confusion set	The sides don't mirror each other.
D	Adjacent category	Radial spreads from a center point.

8. Answer D Concept 6

In a banking app prototype ~~with a navy theme~~, the Next button is green on page 1, blue on page 2, and a plain text link on page 3. Users hesitate at each step. ~~The bank was founded in 1920.~~ Which principle is being broken?

Target word: broken principle **Decides:** The same action looks different each time.

A	Inverted operation	More variety makes it worse.
B	Adjacent category	Balance is about visual weight on a page.
C	Camouflage	Not a usability principle.
D	Correct	The same control should look the same everywhere.

9. Answer B Concept 7

During a usability test of a recipe app from a startup with 12 employees, the facilitator says, “Click the green Save button to save this recipe.” ~~The office has a test kitchen.~~ What is the problem?

Target word: testing flaw **Decides:** The facilitator gave the answer.

A	Camouflage	Location isn’t the flaw.
B	Correct	Leading hides whether users can find the button.
C	Dropped element	Sample size isn’t described as the issue.
D	Absolute language	They give goals; they just shouldn’t give answers.

10. Answer C Concept 7 • transfer

In a usability test, all five participants fail to find the settings menu. What does this indicate?

Target word: interpret results **Decides:** Repeated failure across users.

A	Wrong stem number	Five users often reveal most major problems.
B	Inverted operation	Blames users for a design issue.
C	Correct	When most users fail, the design is at fault.
D	Absolute language	Ignores clear evidence.

11. Answer B Concept 8

A meal-kit company that ships to 30 states puts its Cancel option in tiny gray text three menus deep, while “Stay subscribed” is a large bright button. What is the issue?

Target word: name the issue **Decides:** Design steers users against their intent.

A	Camouflage	Irrelevant.
B	Correct	Deceptive design is unethical and erodes trust.
C	Confusion set	The issue is manipulation, not mismatched styles.
D	Inverted operation	Hierarchy should serve users, not trap them.

12. Answer B Concept 8 • transfer

WCAG 2.1 AA requires a contrast ratio of at least 4.5:1 for normal-size text. A designer measures gray body text on white at 3.2:1. Which statement is correct?

Target word: pass or fail **Decides:** 3.2 is below 4.5.

A	Confusion set	3:1 is the large-text threshold.
B	Correct	3.2:1 is below the 4.5:1 minimum.
C	Adjacent category	7:1 is the AAA level.
D	Absolute language	Visibility to some users isn’t the standard.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	1	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Storyboards	A storyboard shows a user's experience as a sequence of scenes in context, including emotions and pain points.	Using a site map or mockup to show a user's story.	Showing how a person experiences a task over time.
2	Wireframes	A wireframe shows layout, structure, and content placement, not final colors, images, or code.	Calling a polished mockup a wireframe.	What a wireframe should contain.
3	Prototype fidelity	Use low fidelity early to test concepts and flows; use high fidelity late to test visual details and interactions.	Polishing before the flow is validated.	Project stage and what needs testing.
4	Visual hierarchy	Size, contrast, color, and position direct attention to what matters most.	Making the key action small and low-contrast.	Users miss an important element.
5	Balance	Symmetrical balance mirrors both sides; asymmetrical balance offsets unequal elements by visual weight.	Calling any uneven layout unbalanced.	One large element vs. several small ones.
6	Consistency	The same action should look and behave the same everywhere.	Changing styles for the same control across screens.	Users hesitate at a familiar action.
7	Usability testing	Give goals, not instructions, observe with think-aloud, and treat repeated failures as design problems.	Leading participants or blaming users.	Facilitator wording or repeated failures.
8	Ethical and accessible design	Avoid dark patterns that trick users; meet WCAG AA contrast of at least 4.5:1 for normal text (3:1 for large text).	Calling manipulative emphasis good hierarchy, or misremembering thresholds.	Hidden cancel options or contrast ratios.

Course scope (public description): Design Applications guides students in applying the knowledge and skills gained in Foundations of Design. Students will gain valuable experience in using storyboards to formulate the visual representation of design concepts. The design phase will then evolve into designing a wireframe and low-fidelity prototype using industry-standard visual design tools. Prototype designs will focus on visual composition, interactivity, and usability. Key user-centered design principles will be applied in balance, thematic consistency, visual aesthetics, and ethical considerations. This course provides a hands-on application to the visual and functional design process, producing industry artifacts for students' professional portfolios.